

Active Antiresonant Hollow-Core Fibres Enabled by Internal Semiconductor Quantum-Dot Coatings

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Abstract We demonstrate an active antiresonant hollow-core fibre with optical gain provided by thin internal semiconductor quantum-dot coatings. In a side-pumped all-fibre cavity, it shows 625 nm laser emission while preserving air-core guidance and eliminating the need for liquid or gas-based gain media.

Introduction

Antiresonant hollow-core fibres (AR-HCFs) are emerging as a highly promising fibre platform, combining ultra-low attenuation, low nonlinearity, reduced latency, and the potential for broader usable bandwidth than conventional solid-core fibres [1]. These properties have made hollow-core fibres increasingly attractive not only for high-power laser delivery and sensing [2, 3], but also for future communication systems in which nonlinear penalties and transmission bandwidth are key constraints [4].

However, despite this strong momentum, HCFs still operate mainly as passive transmission media. Active optical gain is currently introduced either by interfacing them with conventional solid-core amplifiers, or by filling the hollow core with gas- or liquid-based gain media [5–7]. While versatile, these approaches typically require external reservoirs, recirculation systems, or sealing concepts that are not naturally compatible with compact fibre-device integration, highlighting a fundamental mismatch between the intrinsic advantages of hollow-core guidance and the need to incorporate bulk gain media. The key unresolved challenge is therefore to introduce optical gain directly into a hollow-core fibre while preserving air-guided propagation.

Colloidal quantum dots (QDs) provide an attractive route to address this challenge. In contrast to conventional gain materials employed in current fibre technologies, QDs offer exceptionally high pump absorption and gain, enabling light amplification and lasing even in solid thin film devices [8]. Additionally, the emission wavelength of QDs can be tuned continuously by bandgap engineering, unconstrained by the discrete ionic transitions of gases or rare-earth emitters [9]. Recent advances in multi-shell colloidal QDs have further improved their suitability as gain materials by suppressing

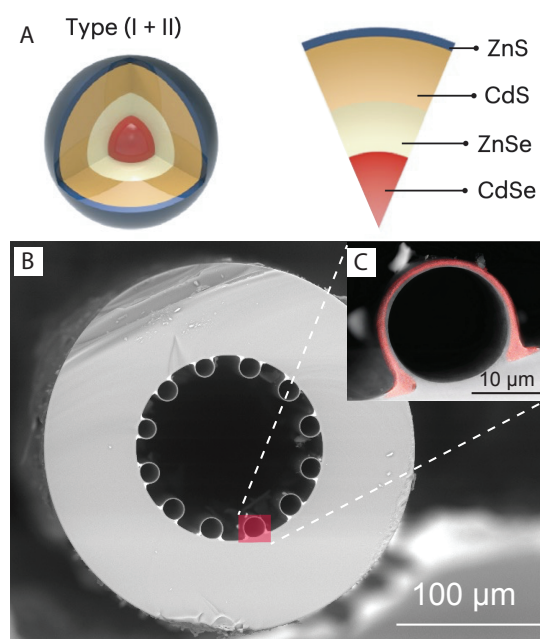


Fig. 1: (A) Schematic cross section of the active CdSe/ZnSe/CdS/ZnS quantum dots. (B) SEM image of the QD-coated AR-HCF. The indicated region is magnified in (C). The QD-coating is highlighted in red.

non-radiative loss channels and enabling efficient lasing in compact cavity geometries [10]. These developments make QDs attractive candidates for non-conventional active fibre device concepts.

Here we report, to the best of our knowledge, the first antiresonant hollow-core fibre device with optical gain provided by a dry internal nanomaterial coating rather than by a gas or liquid core medium. CdSe/ZnSe/CdS/ZnS multishell QDs are deposited on the internal surfaces of a revolver-type AR-HCF, preserving the air-guided character of the structure while introducing gain in the visible spectral range (Fig. 1). Under 532 nm, 10 ns pumping, the device exhibits laser emission from the fibre core near 625 nm in a fibre-coupled cavity.

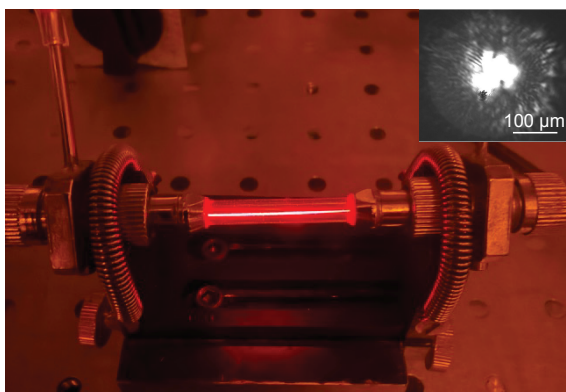


Fig. 2: Photo of the side-pumped laser cavity with Fresnel feedback from solid-core fibre interfaces. The inset shows filtered end-facet emission in longitudinal pumping geometry, confirming core-localized ASE around 625 nm.

This result establishes semiconductor functionalization as a viable route toward active hollow-core fibre devices.

Device Concept

The central device concept is to introduce optical gain into an AR-HCF without replacing the air-guided core by a gas or liquid medium. Instead, semiconductor QDs are deposited as a thin solid coating on the internal microstructure, so that the fibre retains hollow-core guidance while the surrounding interfaces become optically active under external pumping. This is the main conceptual shift that distinguishes the present work from gas or optofluidic hollow-core gain systems investigated in prior art.

Exploiting the extremely high absorption and emission cross sections of modern semiconductor QDs [8], sufficient gain is provided over a device length of only a few centimetres despite the limited overlap between guided mode and QD coating. Once coated, the QD-functionalized AR-HCF no longer requires external filling or recirculation systems and can in principle be directly connected or spliced with other solid- or hollow-core fibres.

Optical gain in colloidal QDs is typically biexcitonic and limited by fast Auger recombination of multicarrier states [8]. To mitigate this, we use a type-(I-II) CdSe/ZnSe/CdS/ZnS multishell QD platform [9], in which advanced heterostructuring suppresses Auger-related loss channels and supports high optical gain with relatively long excited-state lifetimes. While the QDs used here emit around 625 nm, their wavelength can in principle be tuned via particle size and composition.

Experimental Setup

The experiments are carried out with an in-house-fabricated revolver-type AR-HCF with a 100 μm core diameter and a transmission window spanning approximately 500–700 nm [11]. As gain medium we selected QDs with diameter of approx.

13 nm employing a core/multishell architecture consisting of a CdSe core surrounded by ZnSe and CdS shells, with a final ZnS layer for improved stability (Fig. 1 (A)). This QD platform provides a gain bandwidth approaching 100 nm at visible wavelengths, a gain lifetime of up to 3 ns, and peak material gain exceeding 1000 cm^{-1} [10].

The QDs are first prepared in toluene at a concentration of 14 $\mu\text{mol/l}$ and then introduced into the fibre using an optofluidic filling system. Residual toluene is subsequently removed by airflow and ambient evaporation, leaving a dry internal semiconductor nanocrystal coating on the surfaces surrounding the hollow core. Scanning electron microscope images confirm the presence of the internal coating (Fig. 1).

The coated AR-HCF is pumped at 532 nm with 10-ns pulses at 10 Hz in two complementary configurations. First, a side-pumped geometry is used. A cylindrical lens forms an approximately 5 cm line focus along the AR-HCF. Passive solid-core multimode fibres connected to both ends provide signal collection and Fresnel reflections that define a simple linear cavity (Fig. 2). This arrangement directly tests whether the internally functionalized AR-HCF can operate as an active device with minimal additional cavity components.

In the second configuration, the fibre is pumped longitudinally without cavity feedback. In this case, forward-emitted amplified spontaneous emission (ASE) is collected after pump rejection with a dichroic mirror. This arrangement is used to verify that the light emission originates from the fibre core and not from parasitic fluorescence in the cladding or surrounding setup.

Results and Discussion

In the side-pumped cavity configuration, the coated AR-HCF exhibits laser emission centred at 625 nm. The measured input-output characteristic and emission spectra below and above threshold are shown in Fig. 3. A sharp lasing threshold is observed at a pump pulse energy of 6 μJ , followed by an approximately linear increase of output intensity with pump energy. Simultaneously, the emission narrows from a broadband photoluminescence background of about 40 nm width to a 625 nm lasing peak with a linewidth of approximately 4 nm. The relatively broad linewidth is consistent with nanosecond pumping and the absence of wavelength-selective intracavity elements.

In the longitudinal cavity-free configuration, forward ASE is observed with a threshold of approximately 300 μJ . End-facet imaging confirms that the amplified emission originates from the fibre core region as shown in the inset of Fig. 2. The emitted light further exhibits partial coherence, with a coherence length consistent with the measured ASE bandwidth on the order of 10 nm.

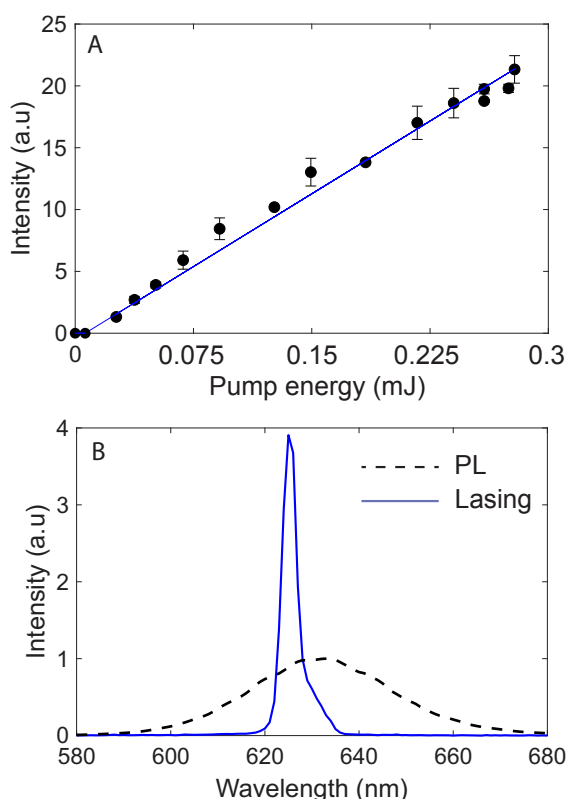


Fig. 3: (A): Output intensity vs. pump energy of the fibre-coupled side-pumped laser cavity. (B) Spectral emission below (PL) and above (lasing) threshold (not to scale).

Discussion

The central significance of this work is the demonstration of optical gain in an AR-HCF via a dry internal semiconductor coating rather than a gas or liquid core medium. The results show that such a coating can provide sufficient gain for active operation while preserving the essential air-guided architecture of the fibre, thereby extending AR-HCFs from a predominantly passive transmission platform toward active hollow-core fibre devices. More broadly, this approach moves hollow-core gain concepts beyond the exclusive use of gas- and liquid-based media, which typically require filling, recirculation, or sealing schemes that are difficult to integrate into compact devices. In contrast, the dry internal coating demonstrated here forms a fixed solid gain layer and is therefore better aligned with splicing and compact system architectures.

A further advantage of the QD platform is its spectral flexibility beyond standard rare-earth-doped fibre platforms. Through bandgap engineering via particle size and composition, the demonstrated operation around 625 nm can in principle be extended toward other visible and near-infrared wavelength ranges, and potentially beyond.

Further work is required for improving coating uniformity, controlling amplified mode quality, and developing pumping schemes better aligned with technologically relevant fibre subsystems.

Conclusion

We have demonstrated an active AR-HCF device enabled by dry internal CdSe/ZnSe/CdS/ZnS quantum-dot coatings. Under 532 nm nanosecond pumping, the coated fibre shows laser emission at 625 nm in a side-pumped all-fibre cavity. This establishes dry semiconductor nanocrystal functionalization as a viable strategy for active hollow-core fibre devices.

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